

3.5.3 Thermal Physics

- **Thermal energy**

Calculations involving change of energy.

For a change of temperature; $Q = mc\Delta T$, where c is specific heat capacity.

For a change of state; $Q = ml$, where l is specific latent heat.

- **Ideal gases**

Gas laws as experimental relationships between p , V , T and mass.

Concept of absolute zero of temperature.

Ideal gas equation as $pV = nRT$ for n moles and as $pV = NkT$ for N molecules.

Avogadro constant N_A , molar gas constant R , Boltzmann constant k .

Molar mass and molecular mass.

- **Molecular kinetic theory model**

Explanation of relationships between p , V and T in terms of a simple molecular model.

Assumptions leading to and derivation of

$$pV = \frac{1}{3} N m (c_{\text{rms}})^2$$

Average molecular kinetic energy

$$\frac{1}{2} m (c_{\text{rms}})^2 = \frac{3}{2} kT = \frac{3RT}{2 N_A}$$